

ELEC 576 / COMP 576 — Fall 2026

Assignment 0

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All six tasks were carried out on macOS 15.1 (Apple silicon, arm64) using a fresh Anaconda installation. Every command and every figure reproduced in this report was executed on that machine; the scripts that generate them are included in the GitHub repository linked in Task 6.

1 Python Machine Learning Stack (Anaconda)

Anaconda was installed from the official Anaconda3-2026.07-1-MacOSX-arm64.sh installer into ~/anaconda3, and conda init zsh was run so that conda is available in new shells. The installation provides conda 26.5.3 on Python 3.14.6, with 552 packages in the base environment.

Task 1 — output of conda info

```
active environment : None
user config file : /Users/jsr.ovo._/.condarc
populated config files : /Users/jsr.ovo._/anaconda3/.condarc
                        /Users/jsr.ovo._/anaconda3/condarc.d/anaconda-auth.yml
conda version : 26.5.3
conda-build version : 26.5.0
python version : 3.14.6.final.0
solver : libmamba (default)
virtual packages : __archspec=1=m4
                  __conda=26.5.3=0
                  __osx=15.1=0
                  __unix=0=0
base environment : /Users/jsr.ovo._/anaconda3 (writable)
conda av data dir : /Users/jsr.ovo._/anaconda3/etc/conda
conda av metadata url : None
channel URLs : https://repo.anaconda.com/pkgs/main/osx-arm64
               https://repo.anaconda.com/pkgs/main/noarch
               https://repo.anaconda.com/pkgs/r/osx-arm64
               https://repo.anaconda.com/pkgs/r/noarch
package cache : /Users/jsr.ovo._/anaconda3/pkgs
                /Users/jsr.ovo._/.conda/pkgs
envs directories : /Users/jsr.ovo._/anaconda3/envs
                  /Users/jsr.ovo._/.conda/envs
temporary directory : /var/folders/_d/tstckvp90tn2cjp23w7v85zw0000gn/T
platform : osx-arm64
user-agent : conda/26.5.3 requests/2.34.2 CPython/3.14.6 Darwin/24.1.0 OSX/15.1
            solver/libmamba conda-libmamba-solver/26.7.0 libmambapy/2.3.2 aau/0.8.1 c/
            s/. e/.
UID:GID : 501:20
netrc file : None
offline mode : False
```

As an additional check that the installation works, conda list returns the full package inventory. Its first lines are shown below; the complete listing (552 packages) is saved in the repository as outputs/task1_conda_list.txt.

```
# packages in environment at /Users/jsr.ovo._/anaconda3:
#
# Name                                Version            Build                                Channel
_anaconda_depends                    2026.07            py314_openblas_1
aiobotocore                          3.7.0              py314hca03da5_0
aiodns                              3.6.1              py314hca03da5_0
aiohappyeyeballs                    2.6.2              py314hca03da5_0
aiohttp                             3.14.1             py314haa24f5a_0
aioitertools                        0.12.0             py314hca03da5_0
aiosignal                           1.4.0              py314hca03da5_0
alabaster                           1.0.0              py314hca03da5_0
altair                              6.0.0              py314hca03da5_1
anaconda-anon-usage                  0.8.1              pyhb46e38b_100
anaconda-auth                        0.15.1             py314hca03da5_0
```

2 Transition from MATLAB to Python

Task 2 asks for every Python command in the *Linear Algebra Equivalents* table of *NumPy for MATLAB Users* to be run in IPython. The table has 82 rows, and where a row lists several equivalent spellings (for example `np.ndim(a)` and `a.ndim`) every alternative was executed, for 161 IPython cells in total.

The matrices are my own choice: `a`, `b`, `c`, `d` are 5×5 with entries drawn uniformly from [0,1) and rounded to two decimals, and `v` is a matching 5-vector. Values in [0,1) were chosen deliberately so that the threshold rows (`a > 0.5`) are non-trivial, and a fixed seed (`default_rng(576)`) makes the whole transcript reproducible.

Deviations from the table, and why

Eight groups of rows cannot be run exactly as printed. Each is annotated in place in the transcript, and they are collected here:

Row 28, 30, 76, 77. These assign or sort in place, which would corrupt `a` for every later row. A copy `a_orig` is kept and restored immediately afterwards; the restore is shown in the transcript rather than done silently.

Row 46. The table prints `np.ix_(np.r_[0:9.], np.r_[0:6.])` — a missing closing parenthesis. Float indices also raise `IndexError`, since `np.ix_` requires integer or boolean arrays, so integer ranges are used.

Row 57, 58. The table writes a bare `logical_and`; it is called as `np.logical_and`.

Row 59, 60. Bitwise `&` and `|` are not defined on float arrays, so a separate integer pair `ai`, `bi` is used.

Row 65. The table gives prose (“Solve $a.T x.T = b.T$ instead”) rather than a command; it is written out as `linalg.solve(a.T, b.T).T`.

Row 67, 73. Cholesky and conjugate gradients need a positive definite system, so `spd = a @ a.T + 5*np.eye(5)` is used instead of `a`.

Row 80. `np.ceil` returns a float and `signal.resample` requires an integer sample count, so the result is wrapped in `int(...)`.

Row 82. `a` has no singleton dimensions, so a 1×3×1×4 array is used to show what `squeeze` actually does.

The full IPython transcript follows. It was produced by `task2_linalg.py`, which drives a real `InteractiveShell` so the `In[ ] / Out[ ]` pairs below are genuine IPython output.

```
#####
# TASK 2 - Linear Algebra Equivalents, NumPy for MATLAB Users
# Every command below was executed in IPython. Matrices are 5x5 with
# entries drawn uniformly from [0,1) and rounded to 2 decimals.
#####

=====
Row 0  SETUP
=====
In [1]: import numpy as np

In [2]: import scipy.linalg

In [3]: from numpy.random import default_rng

In [4]: from scipy import signal

In [5]: from scipy.sparse.linalg import eigs, cg

In [6]: linalg = scipy.linalg          # the assignment asks us to import scipy.linalg

In [7]: np.set_printoptions(precision=4, suppress=True, linewidth=100)

In [8]: seed = default_rng(576)

In [9]: a = np.round(seed.random((5, 5)), 2)

In [10]: b = np.round(seed.random((5, 5)), 2)

In [11]: c = np.round(seed.random((5, 5)), 2)

In [12]: d = np.round(seed.random((5, 5)), 2)

In [13]: v = np.round(seed.random(5), 2)

In [14]: x = a.copy()

In [15]: m, n, q = 2, 2, 2

In [16]: ai = seed.integers(0, 8, (5, 5))          # integer arrays, for the bitwise rows

In [17]: bi = seed.integers(0, 8, (5, 5))

In [18]: spd = a @ a.T + 5 * np.eye(5)            # symmetric positive definite, for chol / cg

In [19]: sig = np.round(seed.random(12), 2)        # a 1-D signal, for resample

In [20]: a_orig = a.copy()                        # kept so the in-place rows can be undone

In [21]: a
Out[21]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12]])

In [22]: b
Out[22]:
array([[0.41, 0.38, 0.82, 0.08, 0.35],
       [0.58, 0.25, 0.54, 0.19, 0.72],
       [0.57, 0.09, 0.79, 0.43, 0.73],
       [0.94, 0.58, 0.54, 0.66, 0.5 ],
       [0.34, 0.41, 0.48, 0.97, 0.23]])

In [23]: v
Out[23]: array([0.66, 0.89, 0.71, 0.42, 0.78])

=====
Row 1  MATLAB: ndims(a)
=====
In [24]: np.ndim(a)
Out[24]: 2

In [25]: a.ndim
Out[25]: 2
```

```

=====
Row 2  MATLAB: numel(a)
=====
In [26]: np.size(a)
Out[26]: 25

In [27]: a.size
Out[27]: 25

=====
Row 3  MATLAB: size(a)
=====
In [28]: np.shape(a)
Out[28]: (5, 5)

In [29]: a.shape
Out[29]: (5, 5)

=====
Row 4  MATLAB: size(a,n)
=====
In [30]: a.shape[n-1]
Out[30]: 5

=====
Row 5  MATLAB: [ 1 2 3; 4 5 6 ]
=====
In [31]: np.array([[1., 2., 3.], [4., 5., 6.]])
Out[31]:
      array([[1., 2., 3.],
             [4., 5., 6.]])

=====
Row 6  MATLAB: [ a b; c d ]
=====
In [32]: np.block([[a, b], [c, d]])
Out[32]:
      array([[0.81, 0.86, 0.94, 0.06, 0.4 , 0.41, 0.38, 0.82, 0.08, 0.35],
             [0.88, 0.3 , 0.44, 0.31, 0.28, 0.58, 0.25, 0.54, 0.19, 0.72],
             [0.54, 0.82, 0.43, 0.28, 0.86, 0.57, 0.09, 0.79, 0.43, 0.73],
             [0.54, 0.58, 0.88, 0.78, 0.21, 0.94, 0.58, 0.54, 0.66, 0.5 ],
             [0.2 , 0.8 , 0.66, 0.28, 0.12, 0.34, 0.41, 0.48, 0.97, 0.23],
             [0.71, 0.34, 0.46, 0.66, 0.89, 0.56, 0.76, 0.74, 0.42, 0.1 ],
             [0.28, 0.9 , 0.44, 0.87, 0.99, 0.39, 0.6 , 0.35, 0.86, 0.28],
             [0.7 , 0.9 , 0.72, 0.65, 0.82, 0.87, 0.32, 0.15, 0.86, 0.04],
             [0.06, 0.86, 0.66, 0.54, 0.1 , 0.35, 0.28, 0.72, 0.04, 0.58],
             [0.45, 0.71, 0.62, 0.82, 0.88, 0.94, 0.17, 0.56, 0.6 , 0.25]])

=====
Row 7  MATLAB: a(end)
=====
In [33]: a[-1]
Out[33]: array([0.2 , 0.8 , 0.66, 0.28, 0.12])

=====
Row 8  MATLAB: a(2,5)
=====
In [34]: a[1, 4]
Out[34]: np.float64(0.28)

=====
Row 9  MATLAB: a(2,:)
=====
In [35]: a[1]
Out[35]: array([0.88, 0.3 , 0.44, 0.31, 0.28])

In [36]: a[1, :]
Out[36]: array([0.88, 0.3 , 0.44, 0.31, 0.28])

=====
Row 10 MATLAB: a(1:5,:)
=====
In [37]: a[0:5]
Out[37]:
      array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
             [0.88, 0.3 , 0.44, 0.31, 0.28],
             [0.54, 0.82, 0.43, 0.28, 0.86],

```

```

        [0.54, 0.58, 0.88, 0.78, 0.21],
        [0.2 , 0.8 , 0.66, 0.28, 0.12]])

In [38]: a[:5]
Out[38]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12]])

In [39]: a[0:5, :]
Out[39]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12]])

=====
Row 11  MATLAB: a(end-4:end,:)
=====
In [40]: a[-5:]
Out[40]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12]])

=====
Row 12  MATLAB: a(1:3,5:9)
NOTE:    a is 5x5, so columns 5..9 resolve to the single column 5.
=====
In [41]: a[0:3, 4:9]
Out[41]:
array([[0.4 ],
       [0.28],
       [0.86]])

=====
Row 13  MATLAB: a([2,4,5],[1,3])
=====
In [42]: a[np.ix_([1, 3, 4], [0, 2])]
Out[42]:
array([[0.88, 0.44],
       [0.54, 0.88],
       [0.2 , 0.66]])

=====
Row 14  MATLAB: a(3:2:21,:)
=====
In [43]: a[2:21:2, :]
Out[43]:
array([[0.54, 0.82, 0.43, 0.28, 0.86],
       [0.2 , 0.8 , 0.66, 0.28, 0.12]])

=====
Row 15  MATLAB: a(1:2:end,:)
=====
In [44]: a[:,2, :]
Out[44]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.2 , 0.8 , 0.66, 0.28, 0.12]])

=====
Row 16  MATLAB: a(end:-1:1,:) or flipud(a)
=====
In [45]: a[::-1, :]
Out[45]:
array([[0.2 , 0.8 , 0.66, 0.28, 0.12],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.81, 0.86, 0.94, 0.06, 0.4 ]])

```

```

=====
Row 17  MATLAB:  a([1:end 1], :)
=====
In [46]: a[np.r_[:len(a), 0]]
Out[46]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12],
       [0.81, 0.86, 0.94, 0.06, 0.4 ]])

=====
Row 18  MATLAB:  a.'
=====
In [47]: a.transpose()
Out[47]:
array([[0.81, 0.88, 0.54, 0.54, 0.2 ],
       [0.86, 0.3 , 0.82, 0.58, 0.8 ],
       [0.94, 0.44, 0.43, 0.88, 0.66],
       [0.06, 0.31, 0.28, 0.78, 0.28],
       [0.4 , 0.28, 0.86, 0.21, 0.12]])

In [48]: a.T
Out[48]:
array([[0.81, 0.88, 0.54, 0.54, 0.2 ],
       [0.86, 0.3 , 0.82, 0.58, 0.8 ],
       [0.94, 0.44, 0.43, 0.88, 0.66],
       [0.06, 0.31, 0.28, 0.78, 0.28],
       [0.4 , 0.28, 0.86, 0.21, 0.12]])

=====
Row 19  MATLAB:  a'
=====
In [49]: a.conj().transpose()
Out[49]:
array([[0.81, 0.88, 0.54, 0.54, 0.2 ],
       [0.86, 0.3 , 0.82, 0.58, 0.8 ],
       [0.94, 0.44, 0.43, 0.88, 0.66],
       [0.06, 0.31, 0.28, 0.78, 0.28],
       [0.4 , 0.28, 0.86, 0.21, 0.12]])

In [50]: a.conj().T
Out[50]:
array([[0.81, 0.88, 0.54, 0.54, 0.2 ],
       [0.86, 0.3 , 0.82, 0.58, 0.8 ],
       [0.94, 0.44, 0.43, 0.88, 0.66],
       [0.06, 0.31, 0.28, 0.78, 0.28],
       [0.4 , 0.28, 0.86, 0.21, 0.12]])

=====
Row 20  MATLAB:  a * b
=====
In [51]: a @ b
Out[51]:
array([[1.5591, 0.8062, 2.0956, 1.06 , 1.7109],
       [1.1722, 0.7436, 1.533 , 0.7928, 1.0646],
       [1.4977, 0.9639, 1.7893, 1.4029, 1.4311],
       [1.864 , 0.9679, 1.9732, 1.2503, 1.6873],
       [1.2262, 0.547 , 1.3262, 0.753 , 1.2954]])

=====
Row 21  MATLAB:  a .* b
=====
In [52]: a * b
Out[52]:
array([[0.3321, 0.3268, 0.7708, 0.0048, 0.14 ],
       [0.5104, 0.075 , 0.2376, 0.0589, 0.2016],
       [0.3078, 0.0738, 0.3397, 0.1204, 0.6278],
       [0.5076, 0.3364, 0.4752, 0.5148, 0.105 ],
       [0.068 , 0.328 , 0.3168, 0.2716, 0.0276]])

=====
Row 22  MATLAB:  a./b
=====
In [53]: a/b
Out[53]:

```

```

array([[1.9756, 2.2632, 1.1463, 0.75 , 1.1429],
       [1.5172, 1.2 , 0.8148, 1.6316, 0.3889],
       [0.9474, 9.1111, 0.5443, 0.6512, 1.1781],
       [0.5745, 1. , 1.6296, 1.1818, 0.42 ],
       [0.5882, 1.9512, 1.375 , 0.2887, 0.5217]])

=====
Row 23  MATLAB:  a.^3
=====
In [54]: a**3
Out[54]:
array([[0.5314, 0.6361, 0.8306, 0.0002, 0.064 ],
       [0.6815, 0.027 , 0.0852, 0.0298, 0.022 ],
       [0.1575, 0.5514, 0.0795, 0.022 , 0.6361],
       [0.1575, 0.1951, 0.6815, 0.4746, 0.0093],
       [0.008 , 0.512 , 0.2875, 0.022 , 0.0017]])

=====
Row 24  MATLAB:  (a > 0.5)
=====
In [55]: (a > 0.5)
Out[55]:
array([[ True,  True,  True, False, False],
       [ True, False, False, False, False],
       [ True,  True, False, False,  True],
       [ True,  True,  True,  True, False],
       [False,  True,  True, False, False]])

=====
Row 25  MATLAB:  find(a > 0.5)
=====
In [56]: np.nonzero(a > 0.5)
Out[56]:
(array([0, 0, 0, 1, 2, 2, 2, 3, 3, 3, 3, 4, 4]),
 array([0, 1, 2, 0, 0, 1, 4, 0, 1, 2, 3, 1, 2]))

=====
Row 26  MATLAB:  a[:,find(v > 0.5)]
=====
In [57]: a[:, np.nonzero(v > 0.5)[0]]
Out[57]:
array([[0.81, 0.86, 0.94, 0.4 ],
       [0.88, 0.3 , 0.44, 0.28],
       [0.54, 0.82, 0.43, 0.86],
       [0.54, 0.58, 0.88, 0.21],
       [0.2 , 0.8 , 0.66, 0.12]])

=====
Row 27  MATLAB:  a[:,find(v>0.5)]
=====
In [58]: a[:, v.T > 0.5]
Out[58]:
array([[0.81, 0.86, 0.94, 0.4 ],
       [0.88, 0.3 , 0.44, 0.28],
       [0.54, 0.82, 0.43, 0.86],
       [0.54, 0.58, 0.88, 0.21],
       [0.2 , 0.8 , 0.66, 0.12]])

=====
Row 28  MATLAB:  a(a<0.5)=0
NOTE:    This row assigns in place, so a is restored afterwards.
=====
In [59]: a[a < 0.5] = 0

In [60]: a
Out[60]:
array([[0.81, 0.86, 0.94, 0. , 0. ],
       [0.88, 0. , 0. , 0. , 0. ],
       [0.54, 0.82, 0. , 0. , 0.86],
       [0.54, 0.58, 0.88, 0.78, 0. ],
       [0. , 0.8 , 0.66, 0. , 0. ]])

In [61]: a = a_orig.copy() # undo, so later rows see the original a

=====
Row 29  MATLAB:  a .* (a>0.5)
=====

```

```

In [62]: a * (a > 0.5)
Out[62]:
array([[0.81, 0.86, 0.94, 0. , 0. ],
       [0.88, 0. , 0. , 0. , 0. ],
       [0.54, 0.82, 0. , 0. , 0.86],
       [0.54, 0.58, 0.88, 0.78, 0. ],
       [0. , 0.8 , 0.66, 0. , 0. ]])

=====
Row 30  MATLAB: a(:) = 3
NOTE:   Also in place; a is restored afterwards.
=====
In [63]: a[:] = 3

In [64]: a
Out[64]:
array([[3., 3., 3., 3., 3.],
       [3., 3., 3., 3., 3.],
       [3., 3., 3., 3., 3.],
       [3., 3., 3., 3., 3.],
       [3., 3., 3., 3., 3.]])

In [65]: a = a_orig.copy() # undo again

=====
Row 31  MATLAB: y=x
=====
In [66]: y = x.copy()

In [67]: y
Out[67]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12]])

=====
Row 32  MATLAB: y=x(2,:)
=====
In [68]: y = x[1, :].copy()

In [69]: y
Out[69]: array([0.88, 0.3 , 0.44, 0.31, 0.28])

=====
Row 33  MATLAB: y=x(:)
=====
In [70]: y = x.flatten()

In [71]: y
Out[71]:
array([0.81, 0.86, 0.94, 0.06, 0.4 , 0.88, 0.3 , 0.44, 0.31, 0.28, 0.54, 0.82, 0.43, 0.28, 0.86,
       0.54, 0.58, 0.88, 0.78, 0.21, 0.2 , 0.8 , 0.66, 0.28, 0.12])

=====
Row 34  MATLAB: 1:10
=====
In [72]: np.arange(1., 11.)
Out[72]: array([ 1.,  2.,  3.,  4.,  5.,  6.,  7.,  8.,  9., 10.])

In [73]: np.r_[1.:11.]
Out[73]: array([ 1.,  2.,  3.,  4.,  5.,  6.,  7.,  8.,  9., 10.])

In [74]: np.r_[1:10:10j]
Out[74]: array([ 1.,  2.,  3.,  4.,  5.,  6.,  7.,  8.,  9., 10.])

=====
Row 35  MATLAB: 0:9
=====
In [75]: np.arange(10.)
Out[75]: array([0., 1., 2., 3., 4., 5., 6., 7., 8., 9.])

In [76]: np.r_[0:10.]
Out[76]: array([0., 1., 2., 3., 4., 5., 6., 7., 8., 9.])

In [77]: np.r_[0:10j]

```

```
Out[77]: array([0., 1., 2., 3., 4., 5., 6., 7., 8., 9.])
```

```
=====  
Row 36  MATLAB:  [1:10]'
```

```
=====  
In [78]: np.arange(1., 11.)[:, np.newaxis]
```

```
Out[78]:  
array([[ 1.],  
       [ 2.],  
       [ 3.],  
       [ 4.],  
       [ 5.],  
       [ 6.],  
       [ 7.],  
       [ 8.],  
       [ 9.],  
       [10.]])
```

```
=====  
Row 37  MATLAB:  zeros(3,4)
```

```
=====  
In [79]: np.zeros((3, 4))
```

```
Out[79]:  
array([[0., 0., 0., 0.],  
       [0., 0., 0., 0.],  
       [0., 0., 0., 0.]])
```

```
=====  
Row 38  MATLAB:  zeros(3,4,5)
```

```
=====  
In [80]: np.zeros((3, 4, 5))
```

```
Out[80]:  
array([[[0., 0., 0., 0., 0.],  
        [0., 0., 0., 0., 0.],  
        [0., 0., 0., 0., 0.],  
        [0., 0., 0., 0., 0.]],  
       [[0., 0., 0., 0., 0.],  
        [0., 0., 0., 0., 0.],  
        [0., 0., 0., 0., 0.],  
        [0., 0., 0., 0., 0.]],  
       [[0., 0., 0., 0., 0.],  
        [0., 0., 0., 0., 0.],  
        [0., 0., 0., 0., 0.],  
        [0., 0., 0., 0., 0.]])
```

```
=====  
Row 39  MATLAB:  ones(3,4)
```

```
=====  
In [81]: np.ones((3, 4))
```

```
Out[81]:  
array([[1., 1., 1., 1.],  
       [1., 1., 1., 1.],  
       [1., 1., 1., 1.]])
```

```
=====  
Row 40  MATLAB:  eye(3)
```

```
=====  
In [82]: np.eye(3)
```

```
Out[82]:  
array([[1., 0., 0.],  
       [0., 1., 0.],  
       [0., 0., 1.]])
```

```
=====  
Row 41  MATLAB:  diag(a)
```

```
=====  
In [83]: np.diag(a)
```

```
Out[83]: array([0.81, 0.3 , 0.43, 0.78, 0.12])
```

```
=====  
Row 42  MATLAB:  diag(v,0)
```

```
=====  
In [84]: np.diag(v, 0)
```

```
Out[84]:  
array([[0.66, 0. , 0. , 0. , 0. ],
```

```

[0. , 0.89, 0. , 0. , 0. ],
[0. , 0. , 0.71, 0. , 0. ],
[0. , 0. , 0. , 0.42, 0. ],
[0. , 0. , 0. , 0. , 0.78]])

=====
Row 43  MATLAB:  rng(42,'twister'); rand(3,4)
=====
In [85]: from numpy.random import default_rng

In [86]: rng = default_rng(42)

In [87]: rng.random((3, 4))
Out[87]: array([[0.774 , 0.4389, 0.8586, 0.6974],
                [0.0942, 0.9756, 0.7611, 0.7861],
                [0.1281, 0.4504, 0.3708, 0.9268]])

=====
Row 44  MATLAB:  linspace(1,3,4)
=====
In [88]: np.linspace(1, 3, 4)
Out[88]: array([1.        , 1.6667, 2.3333, 3.        ])

=====
Row 45  MATLAB:  [x,y]=meshgrid(0:8,0:5)
=====
In [89]: np.mgrid[0:9., 0:6.]
Out[89]: array([[0., 0., 0., 0., 0., 0.],
                [1., 1., 1., 1., 1., 1.],
                [2., 2., 2., 2., 2., 2.],
                [3., 3., 3., 3., 3., 3.],
                [4., 4., 4., 4., 4., 4.],
                [5., 5., 5., 5., 5., 5.],
                [6., 6., 6., 6., 6., 6.],
                [7., 7., 7., 7., 7., 7.],
                [8., 8., 8., 8., 8., 8.]],

                [[0., 1., 2., 3., 4., 5.],
                [0., 1., 2., 3., 4., 5.],
                [0., 1., 2., 3., 4., 5.],
                [0., 1., 2., 3., 4., 5.],
                [0., 1., 2., 3., 4., 5.],
                [0., 1., 2., 3., 4., 5.],
                [0., 1., 2., 3., 4., 5.],
                [0., 1., 2., 3., 4., 5.],
                [0., 1., 2., 3., 4., 5.]])

In [90]: np.meshgrid(np.r_[0:9.], np.r_[0:6.])
Out[90]: (array([[0., 1., 2., 3., 4., 5., 6., 7., 8.],
                [0., 1., 2., 3., 4., 5., 6., 7., 8.],
                [0., 1., 2., 3., 4., 5., 6., 7., 8.],
                [0., 1., 2., 3., 4., 5., 6., 7., 8.],
                [0., 1., 2., 3., 4., 5., 6., 7., 8.],
                [0., 1., 2., 3., 4., 5., 6., 7., 8.]]),
          array([[0., 0., 0., 0., 0., 0., 0., 0., 0.],
                [1., 1., 1., 1., 1., 1., 1., 1., 1.],
                [2., 2., 2., 2., 2., 2., 2., 2., 2.],
                [3., 3., 3., 3., 3., 3., 3., 3., 3.],
                [4., 4., 4., 4., 4., 4., 4., 4., 4.],
                [5., 5., 5., 5., 5., 5., 5., 5., 5.]])

=====
Row 46  MATLAB:  (best way to eval functions on a grid)
NOTE:   The table prints np.ix_(np.r_[0:9.],np.r_[0:6.] - a missing paren, and float indices raise
        IndexError. Integer ranges are used here.
=====
In [91]: np.ogrid[0:9., 0:6.]
Out[91]: (array([[0.],
                [1.],
                [2.],
                [3.],
                [4.],
                [5.],
```

```

        [6.],
        [7.],
        [8.]]),
    array([[0., 1., 2., 3., 4., 5.])))

In [92]: np.ix_(np.r_[0:9], np.r_[0:6])
Out[92]:
(array([[0],
        [1],
        [2],
        [3],
        [4],
        [5],
        [6],
        [7],
        [8]]),
 array([[0, 1, 2, 3, 4, 5]]))

=====
Row 47  MATLAB:  [x,y]=meshgrid([1,2,4],[2,4,5])
=====
In [93]: np.meshgrid([1, 2, 4], [2, 4, 5])
Out[93]:
(array([[1, 2, 4],
        [1, 2, 4],
        [1, 2, 4]]),
 array([[2, 2, 2],
        [4, 4, 4],
        [5, 5, 5]]))

=====
Row 48  MATLAB:  (best way to eval functions on a grid)
=====
In [94]: np.ix_([1, 2, 4], [2, 4, 5])
Out[94]:
(array([[1],
        [2],
        [4]]),
 array([[2, 4, 5]]))

=====
Row 49  MATLAB:  repmat(a, m, n)
=====
In [95]: np.tile(a, (m, n))
Out[95]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 , 0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28, 0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86, 0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21, 0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12, 0.2 , 0.8 , 0.66, 0.28, 0.12],
       [0.81, 0.86, 0.94, 0.06, 0.4 , 0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28, 0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86, 0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21, 0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12, 0.2 , 0.8 , 0.66, 0.28, 0.12]])

=====
Row 50  MATLAB:  [a b]
=====
In [96]: np.concatenate((a, b), 1)
Out[96]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 , 0.41, 0.38, 0.82, 0.08, 0.35],
       [0.88, 0.3 , 0.44, 0.31, 0.28, 0.58, 0.25, 0.54, 0.19, 0.72],
       [0.54, 0.82, 0.43, 0.28, 0.86, 0.57, 0.09, 0.79, 0.43, 0.73],
       [0.54, 0.58, 0.88, 0.78, 0.21, 0.94, 0.58, 0.54, 0.66, 0.5 ],
       [0.2 , 0.8 , 0.66, 0.28, 0.12, 0.34, 0.41, 0.48, 0.97, 0.23]])

In [97]: np.hstack((a, b))
Out[97]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 , 0.41, 0.38, 0.82, 0.08, 0.35],
       [0.88, 0.3 , 0.44, 0.31, 0.28, 0.58, 0.25, 0.54, 0.19, 0.72],
       [0.54, 0.82, 0.43, 0.28, 0.86, 0.57, 0.09, 0.79, 0.43, 0.73],
       [0.54, 0.58, 0.88, 0.78, 0.21, 0.94, 0.58, 0.54, 0.66, 0.5 ],
       [0.2 , 0.8 , 0.66, 0.28, 0.12, 0.34, 0.41, 0.48, 0.97, 0.23]])

In [98]: np.column_stack((a, b))
Out[98]:

```

```

array([[0.81, 0.86, 0.94, 0.06, 0.4 , 0.41, 0.38, 0.82, 0.08, 0.35],
       [0.88, 0.3 , 0.44, 0.31, 0.28, 0.58, 0.25, 0.54, 0.19, 0.72],
       [0.54, 0.82, 0.43, 0.28, 0.86, 0.57, 0.09, 0.79, 0.43, 0.73],
       [0.54, 0.58, 0.88, 0.78, 0.21, 0.94, 0.58, 0.54, 0.66, 0.5 ],
       [0.2 , 0.8 , 0.66, 0.28, 0.12, 0.34, 0.41, 0.48, 0.97, 0.23]])

In [99]: np.c_[a, b]
Out[99]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 , 0.41, 0.38, 0.82, 0.08, 0.35],
       [0.88, 0.3 , 0.44, 0.31, 0.28, 0.58, 0.25, 0.54, 0.19, 0.72],
       [0.54, 0.82, 0.43, 0.28, 0.86, 0.57, 0.09, 0.79, 0.43, 0.73],
       [0.54, 0.58, 0.88, 0.78, 0.21, 0.94, 0.58, 0.54, 0.66, 0.5 ],
       [0.2 , 0.8 , 0.66, 0.28, 0.12, 0.34, 0.41, 0.48, 0.97, 0.23]])

=====
Row 51  MATLAB:  [a; b]
=====
In [100]: np.concatenate((a, b))
Out[100]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12],
       [0.41, 0.38, 0.82, 0.08, 0.35],
       [0.58, 0.25, 0.54, 0.19, 0.72],
       [0.57, 0.09, 0.79, 0.43, 0.73],
       [0.94, 0.58, 0.54, 0.66, 0.5 ],
       [0.34, 0.41, 0.48, 0.97, 0.23]])

In [101]: np.vstack((a, b))
Out[101]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12],
       [0.41, 0.38, 0.82, 0.08, 0.35],
       [0.58, 0.25, 0.54, 0.19, 0.72],
       [0.57, 0.09, 0.79, 0.43, 0.73],
       [0.94, 0.58, 0.54, 0.66, 0.5 ],
       [0.34, 0.41, 0.48, 0.97, 0.23]])

In [102]: np.r_[a, b]
Out[102]:
array([[0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.2 , 0.8 , 0.66, 0.28, 0.12],
       [0.41, 0.38, 0.82, 0.08, 0.35],
       [0.58, 0.25, 0.54, 0.19, 0.72],
       [0.57, 0.09, 0.79, 0.43, 0.73],
       [0.94, 0.58, 0.54, 0.66, 0.5 ],
       [0.34, 0.41, 0.48, 0.97, 0.23]])

=====
Row 52  MATLAB:  max(max(a))
=====
In [103]: a.max()
Out[103]: np.float64(0.94)

In [104]: np.nanmax(a)
Out[104]: np.float64(0.94)

=====
Row 53  MATLAB:  max(a)
=====
In [105]: a.max(0)
Out[105]: array([0.88, 0.86, 0.94, 0.78, 0.86])

=====
Row 54  MATLAB:  max(a,[,2)
=====
In [106]: a.max(1)
Out[106]: array([0.94, 0.88, 0.86, 0.88, 0.8 ])

```

```

=====
Row 55  MATLAB:  max(a,b)
=====
In [107]: np.maximum(a, b)
Out[107]:
array([[0.81, 0.86, 0.94, 0.08, 0.4 ],
       [0.88, 0.3 , 0.54, 0.31, 0.72],
       [0.57, 0.82, 0.79, 0.43, 0.86],
       [0.94, 0.58, 0.88, 0.78, 0.5 ],
       [0.34, 0.8 , 0.66, 0.97, 0.23]])

=====
Row 56  MATLAB:  norm(v)
=====
In [108]: np.sqrt(v @ v)
Out[108]: np.float64(1.586379525838631)

In [109]: np.linalg.norm(v)
Out[109]: np.float64(1.586379525838631)

=====
Row 57  MATLAB:  a & b
NOTE:    The table writes a bare logical_and; it lives in the np namespace.
=====
In [110]: np.logical_and(a, b)
Out[110]:
array([[ True,  True,  True,  True,  True],
       [ True,  True,  True,  True,  True],
       [ True,  True,  True,  True,  True],
       [ True,  True,  True,  True,  True],
       [ True,  True,  True,  True,  True]])

=====
Row 58  MATLAB:  a | b
=====
In [111]: np.logical_or(a, b)
Out[111]:
array([[ True,  True,  True,  True,  True],
       [ True,  True,  True,  True,  True],
       [ True,  True,  True,  True,  True],
       [ True,  True,  True,  True,  True],
       [ True,  True,  True,  True,  True]])

=====
Row 59  MATLAB:  bitand(a,b)
NOTE:    Bitwise operators need integer arrays, so the integer pair ai, bi is used.
=====
In [112]: ai & bi
Out[112]:
array([[2, 2, 0, 0, 0],
       [0, 4, 6, 4, 6],
       [0, 4, 6, 0, 5],
       [2, 0, 0, 0, 4],
       [1, 2, 7, 0, 1]])

=====
Row 60  MATLAB:  bitor(a,b)
NOTE:    Same as row 59.
=====
In [113]: ai | bi
Out[113]:
array([[2, 7, 7, 0, 3],
       [3, 5, 7, 7, 6],
       [1, 7, 6, 7, 7],
       [7, 0, 6, 4, 5],
       [7, 2, 7, 7, 7]])

=====
Row 61  MATLAB:  inv(a)
=====
In [114]: linalg.inv(a)
Out[114]:
array([[-0.5073,  2.2723, -0.3672, -1.0236,  0.8117],
       [-1.5891,  1.7277,  0.1441, -1.8824,  3.5275],
       [ 2.6014, -2.7594, -0.4208,  2.0618, -2.8253],
       [-1.6941,  0.7799,  0.2488,  0.7514,  0.7288],
       [ 1.0846, -1.9483,  1.3854,  1.162 , -2.6978]])

```

```

=====
Row 62  MATLAB:  pinv(a)
=====
In [115]: linalg.pinv(a)
Out[115]:
array([[ -0.5073,  2.2723, -0.3672, -1.0236,  0.8117],
       [ -1.5891,  1.7277,  0.1441, -1.8824,  3.5275],
       [  2.6014, -2.7594, -0.4208,  2.0618, -2.8253],
       [ -1.6941,  0.7799,  0.2488,  0.7514,  0.7288],
       [  1.0846, -1.9483,  1.3854,  1.162 , -2.6978]])

=====
Row 63  MATLAB:  rank(a)
=====
In [116]: np.linalg.matrix_rank(a)
Out[116]: np.int64(5)

=====
Row 64  MATLAB:  a\b
=====
In [117]: linalg.solve(a, b)
Out[117]:
array([[ 0.2144,  0.0813,  0.3578,  0.345 ,  0.8653],
       [-0.1375,  0.1955,  0.4204,  2.4424,  0.663 ],
       [ 0.2038,  0.2983,  0.0679, -1.8768, -1.0023],
       [ 0.8538,  0.3083, -0.0158,  1.3226,  0.6936],
       [ 0.2794, -0.3824,  0.2643, -1.5376, -0.0514]])

In [118]: linalg.lstsq(a, b)
Out[118]:
(array([[ 0.2144,  0.0813,  0.3578,  0.345 ,  0.8653],
       [-0.1375,  0.1955,  0.4204,  2.4424,  0.663 ],
       [ 0.2038,  0.2983,  0.0679, -1.8768, -1.0023],
       [ 0.8538,  0.3083, -0.0158,  1.3226,  0.6936],
       [ 0.2794, -0.3824,  0.2643, -1.5376, -0.0514]]),
 array([], dtype=float64),
 np.int64(5),
 array([2.7835, 0.7063, 0.6235, 0.5049, 0.1221]))

=====
Row 65  MATLAB:  b/a
NOTE:    The table says 'Solve a.T x.T = b.T instead'; that is written out here.
=====
In [119]: linalg.solve(a.T, b.T).T
Out[119]:
array([[ 1.5654, -1.2941,  0.0639,  1.0225, -1.5294],
       [ 1.1723, -0.9949,  0.6406,  1.0285, -1.9769],
       [ 1.6862, -1.8162,  0.5896,  2.0474, -3.1078],
       [-0.5696,  1.1885,  0.3681,  0.1363,  0.4155],
       [-0.9692,  0.4648,  0.2923,  0.866 ,  0.4526]])

=====
Row 66  MATLAB:  [U,S,V]=svd(a)
=====
In [120]: U, S, Vh = linalg.svd(a); V = Vh.T

In [121]: U
Out[121]:
array([[ -0.5446, -0.1601, -0.0456, -0.7099,  0.4144],
       [ -0.3588, -0.0775,  0.764 , -0.0638, -0.5268],
       [ -0.4618, -0.6677, -0.2098,  0.5433,  0.0426],
       [ -0.4796,  0.6603,  0.1312,  0.4256,  0.3682],
       [ -0.3624,  0.2942, -0.5942, -0.1255, -0.6429]])

In [122]: S
Out[122]: array([2.7835, 0.7063, 0.6235, 0.5049, 0.1221])

In [123]: V
Out[123]:
array([[ -0.4806, -0.2025,  0.7604, -0.2636, -0.2834],
       [ -0.5471, -0.1276, -0.6115, -0.0747, -0.5522],
       [ -0.5495,  0.4297, -0.118 , -0.3368,  0.6213],
       [ -0.269 ,  0.5335,  0.1785,  0.7656, -0.158 ],
       [ -0.3088, -0.6881, -0.0457,  0.4747,  0.4513]])

=====
Row 67  MATLAB:  chol(a)
=====

```

```

NOTE: Cholesky needs a positive definite matrix, so  $\text{spd} = \mathbf{a} @ \mathbf{a.T} + 5\mathbf{I}$  is used.
=====
In [124]: linalg.cholesky(spd)
Out[124]:
array([[2.7282, 0.5553, 0.6992, 0.6943, 0.5627],
       [0. , 2.434 , 0.3491, 0.3909, 0.2113],
       [0. , 0. , 2.5211, 0.366 , 0.3023],
       [0. , 0. , 0. , 2.5072, 0.324 ],
       [0. , 0. , 0. , 0. , 2.3771]])

=====
Row 68 MATLAB: [V,D]=eig(a)
=====
In [125]: D, V = linalg.eig(a)

In [126]: D
Out[126]: array([ 2.652 +0.j , -0.6227+0.j , 0.5534+0.j , -0.0713+0.2786j, -0.0713-0.2786j])

In [127]: V
Out[127]:
array([[ 0.5051+0.j , 0.3539+0.j , -0.4778+0.j , -0.1411-0.2991j, -0.1411+0.2991j],
       [ 0.3841+0.j , -0.384 +0.j , -0.1696+0.j , -0.5436+0.1455j, -0.5436-0.1455j],
       [ 0.4611+0.j , -0.485 +0.j , 0.1419+0.j , 0.5901+0.j , 0.5901-0.j ],
       [ 0.5195+0.j , 0.2275+0.j , 0.8228+0.j , -0.2182-0.0436j, -0.2182+0.0436j],
       [ 0.3389+0.j , 0.6636+0.j , 0.2142+0.j , 0.334 +0.2544j, 0.334 -0.2544j]])

=====
Row 69 MATLAB: [V,D]=eig(a,b)
=====
In [128]: D, V = linalg.eig(a, b)

In [129]: D
Out[129]: array([-2.0252+0.j , -1.3202+0.j , 1.5696+0.j , 0.4611+0.4215j, 0.4611-0.4215j])

In [130]: V
Out[130]:
array([[ 0.0841+0.j , -0.1273+0.j , 0.0112+0.j , 0.2637+0.1323j, 0.2637-0.1323j],
       [-0.7583+0.j , -0.7367+0.j , -0.6623+0.j , -0.3473+0.3557j, -0.3473-0.3557j],
       [ 0.5327+0.j , 0.6178+0.j , -0.6383+0.j , 0.0315-0.3647j, 0.0315+0.3647j],
       [ 0.2088+0.j , 0.2082+0.j , -0.1113+0.j , -0.1162+0.4314j, -0.1162-0.4314j],
       [-0.301 +0.j , -0.1263+0.j , 0.3761+0.j , 0.5258-0.236j , 0.5258+0.236j ]])

=====
Row 70 MATLAB: [V,D]=eigs(a,3)
NOTE: eigs comes from scipy.sparse.linalg and needs  $k < n-1$ .
=====
In [131]: D, V = eigs(a, k=3)

In [132]: D
Out[132]: array([ 2.652 +0.j, 0.5534+0.j, -0.6227+0.j])

In [133]: V
Out[133]:
array([[ 0.5051+0.j, 0.4778+0.j, -0.3539+0.j],
       [ 0.3841+0.j, 0.1696+0.j, 0.384 +0.j],
       [ 0.4611+0.j, -0.1419+0.j, 0.485 +0.j],
       [ 0.5195+0.j, -0.8228+0.j, -0.2275+0.j],
       [ 0.3389+0.j, -0.2142+0.j, -0.6636+0.j]])

=====
Row 71 MATLAB: [Q,R]=qr(a,0)
=====
In [134]: Q, R = linalg.qr(a)

In [135]: Q
Out[135]:
array([[ -0.5652, 0.1372, 0.1677, 0.7469, -0.2753],
       [ -0.6141, -0.5769, -0.1607, -0.1404, 0.4946],
       [ -0.3768, 0.3737, -0.6986, -0.3266, -0.3517],
       [ -0.3768, 0.0991, 0.6674, -0.5619, -0.295 ],
       [ -0.1396, 0.7063, 0.1122, -0.0081, 0.6848]])

In [136]: R
Out[136]:
array([[ -1.4331, -1.3095, -1.3872, -0.6628, -0.818 ],
       [ 0. , 0.8739, 0.5892, 0.2091, 0.3203],
       [ 0. , 0. , 0.4479, 0.3166, -0.4251],
```

```

[ 0. , 0. , 0. , -0.5307, -0.1404],
[ 0. , 0. , 0. , 0. , -0.2538]])

=====
Row 72  MATLAB: [L,U,P]=lu(a)
=====
In [137]: P, L, U = linalg.lu(a)

In [138]: P
Out[138]:
array([[0., 0., 0., 1., 0.],
       [1., 0., 0., 0., 0.],
       [0., 0., 1., 0., 0.],
       [0., 0., 0., 0., 1.],
       [0., 1., 0., 0., 0.]])

In [139]: L
Out[139]:
array([[ 1. , 0. , 0. , 0. , 0. ],
       [ 0.2273, 1. , 0. , 0. , 0. ],
       [ 0.6136, 0.8689, 1. , 0. , 0. ],
       [ 0.9205, 0.7978, -0.2701, 1. , 0. ],
       [ 0.6136, 0.541 , -0.9401, -0.9333, 1. ]])

In [140]: U
Out[140]:
array([[ 0.88 , 0.3 , 0.44 , 0.31 , 0.28 ],
       [ 0. , 0.7318, 0.56 , 0.2095, 0.0564],
       [ 0. , 0. , -0.3266, -0.0923, 0.6392],
       [ 0. , 0. , 0. , -0.4175, 0.27 ],
       [ 0. , 0. , 0. , 0. , 0.8606]])

In [141]: np.allclose(a, P @ L @ U)
Out[141]: True

=====
Row 73  MATLAB: conjgrad
NOTE: cg is scipy's conjugate gradient solver; it needs an SPD system.
=====
In [142]: cg(spd, v)
Out[142]: (array([ 0.0327, 0.1116, 0.0572, -0.001 , 0.0916]), 0)

=====
Row 74  MATLAB: fft(a)
=====
In [143]: np.fft.fft(a)
Out[143]:
array([[ 3.07 +0.j , 0.3903-0.9547j, 0.0997+0.5665j, 0.0997-0.5665j, 0.3903+0.9547j],
       [ 2.21 +0.j , 0.4525-0.0954j, 0.6425+0.1119j, 0.6425-0.1119j, 0.4525+0.0954j],
       [ 2.93 +0.j , 0.4847-0.0501j, -0.5997+0.1662j, -0.5997-0.1662j, 0.4847+0.0501j],
       [ 2.99 +0.j , -0.5588-0.4107j, 0.4138-0.1224j, 0.4138+0.1224j, -0.5588+0.4107j],
       [ 2.06 +0.j , -0.2762-0.8701j, -0.2538-0.0383j, -0.2538+0.0383j, -0.2762+0.8701j]])

=====
Row 75  MATLAB: ifft(a)
=====
In [144]: np.fft.ifft(a)
Out[144]:
array([[ 0.614 +0.j , 0.0781+0.1909j, 0.0199-0.1133j, 0.0199+0.1133j, 0.0781-0.1909j],
       [ 0.442 +0.j , 0.0905+0.0191j, 0.1285-0.0224j, 0.1285+0.0224j, 0.0905-0.0191j],
       [ 0.586 +0.j , 0.0969+0.01j , -0.1199-0.0332j, -0.1199+0.0332j, 0.0969-0.01j ],
       [ 0.598 +0.j , -0.1118+0.0821j, 0.0828+0.0245j, 0.0828-0.0245j, -0.1118-0.0821j],
       [ 0.412 +0.j , -0.0552+0.174j , -0.0508+0.0077j, -0.0508-0.0077j, -0.0552-0.174j ]])

=====
Row 76  MATLAB: sort(a)
=====
In [145]: np.sort(a)
Out[145]:
array([[0.06, 0.4 , 0.81, 0.86, 0.94],
       [0.28, 0.3 , 0.31, 0.44, 0.88],
       [0.28, 0.43, 0.54, 0.82, 0.86],
       [0.21, 0.54, 0.58, 0.78, 0.88],
       [0.12, 0.2 , 0.28, 0.66, 0.8 ]])

In [146]: a.sort(axis=0); a
Out[146]:

```

```

array([[0.2 , 0.3 , 0.43, 0.06, 0.12],
       [0.54, 0.58, 0.44, 0.28, 0.21],
       [0.54, 0.8 , 0.66, 0.28, 0.28],
       [0.81, 0.82, 0.88, 0.31, 0.4 ],
       [0.88, 0.86, 0.94, 0.78, 0.86]])

In [147]: a = a_orig.copy() # sort is in place

=====
Row 77  MATLAB: sort(a, 2)
=====
In [148]: np.sort(a, axis=1)
Out[148]:
array([[0.06, 0.4 , 0.81, 0.86, 0.94],
       [0.28, 0.3 , 0.31, 0.44, 0.88],
       [0.28, 0.43, 0.54, 0.82, 0.86],
       [0.21, 0.54, 0.58, 0.78, 0.88],
       [0.12, 0.2 , 0.28, 0.66, 0.8 ]])

In [149]: a.sort(axis=1); a
Out[149]:
array([[0.06, 0.4 , 0.81, 0.86, 0.94],
       [0.28, 0.3 , 0.31, 0.44, 0.88],
       [0.28, 0.43, 0.54, 0.82, 0.86],
       [0.21, 0.54, 0.58, 0.78, 0.88],
       [0.12, 0.2 , 0.28, 0.66, 0.8 ]])

In [150]: a = a_orig.copy()

=====
Row 78  MATLAB: [b,I]=sortrows(a,1)
=====
In [151]: I = np.argsort(a[:, 0]); b = a[I, :]

In [152]: I
Out[152]: array([4, 2, 3, 0, 1])

In [153]: b
Out[153]:
array([[0.2 , 0.8 , 0.66, 0.28, 0.12],
       [0.54, 0.82, 0.43, 0.28, 0.86],
       [0.54, 0.58, 0.88, 0.78, 0.21],
       [0.81, 0.86, 0.94, 0.06, 0.4 ],
       [0.88, 0.3 , 0.44, 0.31, 0.28]])

=====
Row 79  MATLAB: x = Z\y
=====
In [154]: Z = a; y = v # rebind, rows 31-33 reassigned x and y

In [155]: x = linalg.lstsq(Z, y)

In [156]: x
Out[156]:
(array([ 1.63 , 2.552 , -2.3754, 0.6368, -1.6508]),
 array([], dtype=float64),
 np.int64(5),
 array([2.7835, 0.7063, 0.6235, 0.5049, 0.1221]))

=====
Row 80  MATLAB: decimate(x, q)
NOTE: np.ceil returns a float; resample needs an int, so int(...) is applied.
=====
In [157]: sig
Out[157]: array([0.1 , 0.97, 0.58, 0.35, 0.15, 0.62, 0.88, 0.44, 0.48, 0.32, 0.47, 0.98])

In [158]: signal.resample(sig, int(np.ceil(len(sig)/q)))
Out[158]: array([0.5243, 0.6842, 0.1882, 0.8574, 0.2725, 0.6435])

=====
Row 81  MATLAB: unique(a)
=====
In [159]: np.unique(a)
Out[159]:
array([0.06, 0.12, 0.2 , 0.21, 0.28, 0.3 , 0.31, 0.4 , 0.43, 0.44, 0.54, 0.58, 0.66, 0.78, 0.8 ,
       0.81, 0.82, 0.86, 0.88, 0.94])

```

```

=====
Row 82  MATLAB:  squeeze(a)
          NOTE:   a has no singleton dimensions, so a 1x3x1x4 array is used to show the effect.
=====
In [160]: np.zeros((1, 3, 1, 4)).squeeze()
Out[160]:
          array([[0., 0., 0., 0.],
                 [0., 0., 0., 0.],
                 [0., 0., 0., 0.]])

In [161]: np.zeros((1, 3, 1, 4)).squeeze().shape
Out[161]: (3, 4)

#####
# End of Task 2. 161 IPython cells executed, all 82 table rows covered.
#####

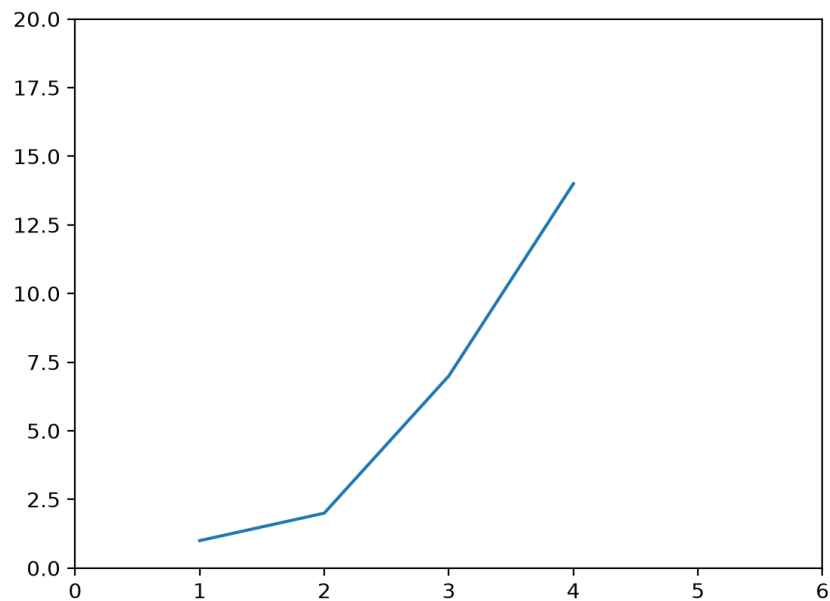
```

3 Plotting — the given script

Task 3 — the assignment's script, run verbatim in IPython

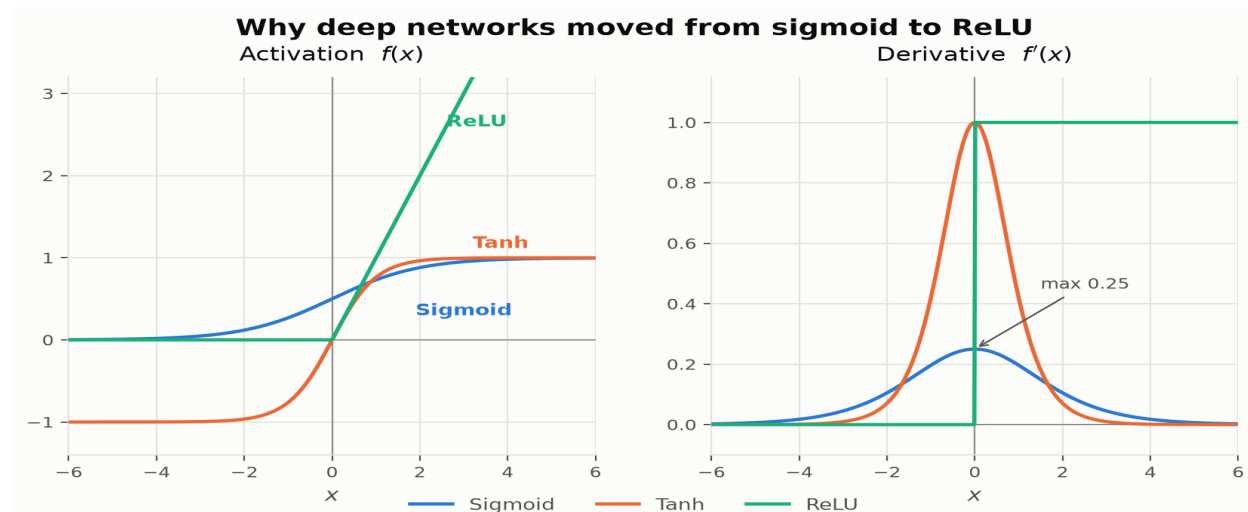
```
import matplotlib.pyplot as plt
plt.plot([1,2,3,4], [1,2,7,14])
plt.axis([0, 6, 0, 20])
plt.show()
```

The figure it produces is below. To capture it to a file the script was run under the non-interactive **Agg** backend and followed by `plt.savefig(...)`; the four lines above are otherwise untouched. Under **Agg**, `plt.show()` emits a `UserWarning` that it cannot display interactively, which is expected and does not affect the figure.



4 Plotting — a figure of my own

For Task 4 I plotted the three classical activation functions against their derivatives, which is the clearest single picture of why deep networks moved away from sigmoid. The left panel shows the functions; the right panel shows the gradients that backpropagation actually multiplies together. Sigmoid's derivative peaks at 0.25 and decays to zero in both tails, so a product of many such factors vanishes exponentially with depth. ReLU's derivative is exactly 1 on the positive half-line, so the gradient passes through unattenuated.



Code

```
"""
ELEC 576 / COMP 576 - Assignment 0, Task 4
A figure of my own: three activation functions and their derivatives.

The right-hand panel is the point of the figure - sigmoid's gradient peaks at
0.25 and decays to zero in both tails, which is the vanishing-gradient problem
that motivates ReLU in deep networks.
"""
import matplotlib
matplotlib.use("Agg")
import numpy as np
import matplotlib.pyplot as plt

SURFACE = "#fcfcfb"
INK = "#0b0b0b"
MUTED = "#52514e"
GRID = "#e4e3df"
SERIES = ["#2a78d6", "#eb6834", "#1baf7a"] # blue, orange, aqua

x = np.linspace(-6, 6, 600)

sigmoid = 1 / (1 + np.exp(-x))
tanh = np.tanh(x)
relu = np.maximum(0, x)

d_sigmoid = sigmoid * (1 - sigmoid)
d_tanh = 1 - tanh**2
d_relu = (x > 0).astype(float)

names = ["Sigmoid", "Tanh", "ReLU"]
funcs = [sigmoid, tanh, relu]
derivs = [d_sigmoid, d_tanh, d_relu]

fig, axes = plt.subplots(1, 2, figsize=(10, 4.2), facecolor=SURFACE)
fig.subplots_adjust(wspace=0.22)
```

```

for ax, ys, title, ylim in [
    (axes[0], funcs, "Activation  $f(x)$ ", (-1.4, 3.2)),
    (axes[1], derivs, "Derivative  $f'(x)$ ", (-0.1, 1.15)),
]:
    ax.set_facecolor(SURFACE)
    ax.grid(True, color=GRID, linewidth=0.8, zorder=0)
    ax.set_axisbelow(True)
    ax.axhline(0, color=MUTED, linewidth=0.8, alpha=0.6, zorder=1)
    ax.axvline(0, color=MUTED, linewidth=0.8, alpha=0.6, zorder=1)
    for y, colour, name in zip(ys, SERIES, names):
        ax.plot(x, y, color=colour, linewidth=2.0, label=name, zorder=3)
    ax.set_ylim(*ylim)
    ax.set_xlim(-6, 6)
    ax.set_title(title, color=INK, fontsize=12, pad=10)
    ax.set_xlabel(" $x$ ", color=MUTED, fontsize=11)
    ax.tick_params(colors=MUTED, labelsize=9)
    for spine in ("top", "right"):
        ax.spines[spine].set_visible(False)
    for spine in ("left", "bottom"):
        ax.spines[spine].set_color(GRID)

# direct labels, placed where the curves are furthest apart
axes[0].annotate("ReLU", xy=(2.6, 2.6), color=SERIES[2], fontsize=10, weight="bold")
axes[0].annotate("Tanh", xy=(3.2, 1.12), color=SERIES[1], fontsize=10, weight="bold")
axes[0].annotate("Sigmoid", xy=(1.9, 0.30), color=SERIES[0], fontsize=10, weight="bold")

axes[1].annotate("max 0.25", xy=(0, 0.25), xytext=(1.5, 0.45),
                color=MUTED, fontsize=9,
                arrowprops=dict(arrowstyle="->", color=MUTED, linewidth=0.9))

handles, labels = axes[0].get_legend_handles_labels()
fig.legend(handles, labels, loc="lower center", ncol=3, frameon=False,
          fontsize=10, labelcolor=MUTED, bbox_to_anchor=(0.5, -0.04))

fig.suptitle("Why deep networks moved from sigmoid to ReLU",
            color=INK, fontsize=13.5, weight="bold", y=1.0)

fig.savefig("figures/task4.png", dpi=200, bbox_inches="tight", facecolor=SURFACE)
print("saved figures/task4.png")

```

5 Version Control System (GitHub)

Task 5 — my VCS account

```
GitHub username:  limerenceaid
Profile:          https://github.com/limerenceaid
```

The repository submitted for Task 6 is hosted under this account.

6 Integrated Development Environment

The project for this assignment was created and run as a standard Python project against the Anaconda interpreter installed in Task 1, then committed and pushed to GitHub as a public repository.

Task 6 — link to the project

```
https://github.com/limerenceaid/ELEC576-Assignment0
```

The repository contains everything used to produce this report:

```
Assignment0/
  task2_linalg.py      all 82 rows of the Linear Algebra Equivalents table
  task3_plot.py        the assignment's plotting script
  task4_plot.py        the activation-function figure
  make_report.py       builds this PDF from the outputs below
  outputs/
    task1_conda_info.txt conda info
    task1_conda_list.txt conda list, all 552 packages
    task2_transcript.txt the full IPython transcript
  figures/
    task3.png
    task4.png
```

Sources

NumPy developers, *NumPy for MATLAB Users*, <https://numpy.org/doc/stable/user/numpy-for-matlab-users.html> — the source of the “Linear Algebra Equivalents” table reproduced in Task 2.

The Matplotlib development team, *Pyplot Tutorial*, <https://matplotlib.org/stable/tutorials/pyplot.html> — background for Tasks 3 and 4.

Anaconda Inc., *Installing on macOS*, <https://docs.anaconda.com/anaconda/install/mac-os/> — installation procedure for Task 1.

The Task 3 script is quoted from the assignment handout. All other code in this report is my own.